

ARIA Independence Roadmap — From Wrapper to Sovereign LLM

Strategic anchor · 2026-05-09 · Target: ARIA-LLM v1.0 by Q1 2027

The directive: ARIA must become a **robust independent AI** — not a wrapper around Anthropic, DeepSeek, or any external LLM, but a sovereign system whose model, memory, and judgement are her own. Best-in-class in security and defence due-diligence.

This document is the canonical roadmap for that transition. It maps the current dependency surface honestly, sets phase exit criteria, and names the operator decisions that gate each phase. It is the multi-month strategic anchor that the next sprints execute against.

1. The Strategic Shift in One Paragraph

ARIA today is a defence-DD intelligence platform whose **brain** — knowledge, ledger, neural, RAG, autonomous engine, constitution, persona overlays — is sovereign on Fly.io. Her **mind** — the LLM that produces text — is rented from Anthropic / DeepSeek / Groq. The shift from rented mind to sovereign mind is the path from "highest-quality wrapper in defence-DD" to "the LLM in defence-DD." At the end of this roadmap, ARIA-LLM is a 70-72B parameter open-weight model fine-tuned on ARIA's own captured outputs, defence-DD primary sources, and the 23-clause constitution. It runs on a dedicated GPU host. It does not phone home. It does not have a kill switch outside Arkmurus's control. That is what robust independence means.

2. The Current Dependency Surface (Honest Map)

ARIA depends on external systems in five categories. Two are deep, three are shallow.

2.1 — DEEP DEPENDENCIES (must replace for independence)

System	Used for	Replacement target
Anthropic API	Primary LLM for chat, DD orchestrator, audit-grade output	ARIA-LLM (own fine-tune) by Phase 4
DeepSeek API	Fallback LLM (cost floor)	ARIA-LLM tier-2 instance OR open-weight alt for redundancy
Groq API	Fallback LLM (speed floor)	Self-hosted vLLM speculative-decoding inference

2.2 — SHALLOW DEPENDENCIES (already self-hosted-able / replaceable)

System	Used for	Status
sentence-transformers/all-MiniLM-L6-v2	RAG embeddings	Already downloaded + cached; runs on CPU; 23 MB model. ZERO external dependency at runtime.
chromadb (RAG store)	Vector retrieval	Self-hosted on <code>/data/aria_rag/</code> . ZERO external dependency.
Upstash Redis	Convenience mirror	Mirrors are non-canonical (R-F94/F110 disk-first). Could swap for self-hosted Redis on Fly.
Tesseract OCR	Document OCR	Self-hosted binary; no external service. ZERO external dependency.
OpenSanctions consolidated lists	Sanctions screening	Free CSV download; we cache locally. ZERO runtime dependency once cached.
OpenAlex / Semantic Scholar / CrossRef	Research backends	Free APIs, no contractual lock-in.
Brave Search	Web search	Paid; circuit breaker OPEN today. R-F86 candidate: SearXNG self-hosted alternative.
Google News RSS	News fallback	Free; no auth required.
49 OSINT sources (sweep)	Live intelligence	Free OSINT, distributed across providers; no single vendor exposure.

The shallow dependencies are not the structural problem. The deep dependencies — three external LLM providers — are.

2.3 — Independence definition

ARIA reaches **independence** when: - All chat / DD / autonomous output is produced by a model running on Arkmurus-controlled compute - The model weights are stored on Arkmurus-controlled storage - No external LLM API call is made on the hot path of any user-facing or audit-grade workflow - An air-gap of ARIA — no internet egress for inference — produces identical behaviour to networked operation

Anthropic / DeepSeek / Groq may remain configured as **emergency break-glass fallbacks**, but the steady state is sovereign.

3. The Five-Phase Roadmap

Phase 1 — Tiered LLM Substitution (Months 0-1, immediate)

Goal: Reduce external LLM spend by 60-80% by routing low-stakes calls to local open-weight models. Anthropic stays as top-tier for audit-grade DD and chat; everything else moves down the tiers.

Build: - **R-F86:** Free-search backend (SearXNG self-hosted on Fly machine) + adapter wired into existing search_doctrine path - **R-F87:** Local LLM tier (Ollama or vLLM) running an 8B-class model (Llama 3.1 8B Instruct or Qwen 2.5 7B Instruct) on a Fly machine OR dedicated micro-host - **R-F87a:** Model tiering middleware in `aria_service/llm/router.py` that dispatches each call to the cheapest tier that meets quality: - **Tier 0 (free, local):** Ollama 8B → student loops (self-quiz), research extraction, classification, signal generation, autonomous task tool dispatch - **Tier 1 (free, local but slower):** Local 8B with reasoning prompt → adversarial grading, FATF typology matching, citation audit - **Tier 2 (cheap external):** DeepSeek → DD layers 1-4 (background extraction, document_intelligence, hypothesis validation) - **Tier 3 (current premium):** Groq → Layer 5c commercial coherence + structured output paths - **Tier 4 (most expensive):** Anthropic Claude → final chat output, audit-grade DD orchestrator, constitutional decisions, customer-facing reports - **Corpus-first query inversion:** every chat turn checks the knowledge base + RAG before any LLM call. Memory hits at \$0. - **Prompt caching:** the 23-clause constitution + persona overlay Layer 1-3 cached on Anthropic's prompt-cache API (10% cost on cache hit vs 100% on miss).

Cost target: monthly LLM spend drops from ~£385/month current estimate to ~£90-130/month by end of Phase 1.

Phase 1 exit criteria (all must pass): 1. ≥70% of LLM calls served by local/free providers (measured via `/api/aria/cost/monthly` per-provider breakdown) 2. Composite adversarial pass rate (R-F59 + R-F80) does not drop more than 5 percentage points vs baseline 3. Customer-facing chat (anything that hits `/api/aria/chat`) latency p50 stays under 8s 4. Output harvester (R-F67) accumulates ≥150 high-quality SFT pairs per day after `ARIA_OUTPUT_HARVEST_ENABLED=1` flips

Operator decisions for Phase 1: - Approve Fly.io machine size bump for Ollama (recommended: shared-cpu-4x 4GB for 8B model, ~£25/mo) - Approve OR reject SearXNG self-host (alternative: continue with paid Brave when topped up) - Confirm acceptable degradation profile for student/research loops (1-2s latency increase from local model)

Phase 2 — Training Corpus Accumulation (Months 1-3, parallel to Phase 1)

Goal: Build the 5,000-10,000 high-quality SFT pair corpus that ARIA-LLM will be fine-tuned on.

Build: - **R-F88:** Learning-progress tracker. For each domain (sanctions, ECCN, EUC, FATF typologies, defence markets, etc.), records: when was each fact last refreshed, what's stale, what hasn't been cycled in 90+ days. Surfaces the freshness picture as an operator dashboard. - **R-F89:** Knowledge coverage heatmap by domain. Visual grid of "what does ARIA know" — domain × jurisdiction × confidence. Gaps are visible. Autonomous engine targets gaps preferentially. - **R-F90:** Continuous-update orchestrator. Wraps the existing autonomous engine with a max-staleness contract: every domain must see at least one fresh signal per . Default windows: sanctions=24h, FATF=7d, EUC=30d, weapons systems=14d. - **Corpus curation pipeline:** in addition to harvested chat outputs, ingest: - DOJ FCPA settlements (R-F69 already feeds these in) - SEC enforcement actions (next iteration of R-F69) - UK SFO published cases - France PNF press releases - NATO STANAGs (publicly available Wassenaar / dual-use guidance) - ITAR/EAR commodity classifications (export.gov/BIS) - FATF mutual evaluation reports (per-country DD context) - SIPRI Yearbook arms transfer data (already partly ingested) - UCDP conflict data (already partly ingested) - **Quality scoring:** every harvested chat output already gets a deterministic quality

score (R-F67). Set the SFT-eligible threshold at 0.80 (vs 0.75 default for general harvest) — only the top responses become training data. - **Adversarial pair generation**: every R-F59 / R-F80 attack response pair becomes a DPO training pair (positive: refusal response; negative: would-be-leak response).

Phase 2 exit criteria: 1. $\geq 5,000$ SFT pairs above quality threshold 0.80 2. ≥ 500 DPO preference pairs from adversarial library 3. $\geq 50,000$ RAG-quality defence-DD documents in chromadb 4. Domain coverage heatmap shows $\geq 85\%$ of cells with non-stale facts 5. Constitutional adherence on harvested corpus: $\geq 95\%$ of pairs pass automated 23-clause check

Operator decisions for Phase 2: - Approve `ARIA_OUTPUT_HARVEST_ENABLED=1` after 3-7 days of dry-run threshold validation - Approve corpus-curation budget for paid defence sources (Janes Defence Weekly archive, IHS Markit defence database) — optional; the free corpus alone is sufficient for v0.1 - Approve PII handling policy for harvested chat (R-F67 already redacts at write time; legal review before fine-tune ingest)

Phase 3 — ARIA-LLM v0.1 Fine-Tune (Month 3-4, GPU rental)

Goal: Produce ARIA-LLM v0.1 — a defence-DD specialised model that beats the underlying base model on Arkmurus's evaluation harness.

Base model decision:

Candidate	Pros	Cons
Llama 3.3 70B Instruct	Mature ecosystem, strong English reasoning, Apache-2.0 commercial-friendly licence variant available, well-supported by vLLM	70B = needs 2× A100 80GB minimum for inference at reasonable latency
Qwen 2.5 72B Instruct	Stronger multilingual (Chinese, Arabic, Portuguese — covers Lusophone moat directly), Apache-2.0, slightly better at structured output	Smaller English-only ecosystem, less battle-tested in Western enterprise
Mixtral 8x22B Instruct	Mixture-of-experts → 39B active params at inference; lower cost-per-call	More finicky to fine-tune cleanly; routing instability under domain shift
DeepSeek-V3	Strong reasoning, low cost, open weights	Foreign provenance — concern for defence-DD given Arkmurus's positioning

Recommendation: **Llama 3.3 70B Instruct** for v0.1. The Lusophone-multilingual gap is real but addressable via Qwen-encoder embedding for retrieval + Llama-decoder for generation; switching base in v0.2 is cheap if needed.

Training protocol: - **SFT (Supervised Fine-Tuning)**: 5k-10k harvested pairs + 2k curated defence-DD synthetic pairs + 1k constitutional-compliance pairs (e.g. "what would Clause 7 say about this?" → correct application). LoRA rank 32, 3 epochs. - **DPO (Direct Preference Optimization)**: 500-1000 preference pairs from R-F59 / R-F80 adversarial outputs. $\beta=0.1$, 1 epoch. - **Constitutional**

reinforcement: optional RLHF-style pass with the 23-clause check as the reward signal. Skip if SFT+DPO already hits evaluation targets.

Evaluation harness: - 33-attack adversarial library (R-F59 + R-F80) — pass rate target 90%+ - 500-question defence-DD evaluation set (curated, held out from training) — accuracy target 80%+ - Citation accuracy (R-F78 audit) on 100 generated DD reports — citation_grounded_rate target 0.85+ - Latency: p50 ≤ 4s on 2× A100, p95 ≤ 12s - Calibration: ECE within 8% (vs 14% underconfident today)

Compute budget: - SFT + DPO on 70B model with LoRA: ~£2,500-4,000 one-time (40-60 GPU-hours on A100 × £45-65/hour cloud rental) - Evaluation harness runs: ~£500 - Total: **£3,000-4,500 one-time fine-tune cost** for v0.1

Phase 3 exit criteria: 1. ARIA-LLM v0.1 passes 90% of the adversarial library (vs 81% baseline target on Anthropic Claude today) 2. ARIA-LLM v0.1 matches Anthropic on defence-DD eval set within 5% 3. Hosted inference benchmark: 2× A100 80GB serves 4 concurrent chat sessions at p50 ≤ 4s 4. Constitutional adherence: 23-clause check passes on 95%+ of generated outputs

Operator decisions for Phase 3: - Approve £3,000-4,500 one-time training compute budget - Approve base model choice (Llama 3.3 70B recommended; Qwen 2.5 72B if Lusophone-first weighting matters more) - Approve training-data PII review (legal — irreversible once weights bake) - Approve ARIA-LLM weight licensing posture (Arkmurus-only; Apache-2.0 if eventual open-source decided)

Phase 4 — ARIA-LLM Production Deployment (Month 4-5)

Goal: ARIA-LLM v0.1 serving production traffic. Anthropic demoted to break-glass fallback only.

Infrastructure decision:

Option	Spec	Monthly cost	Pros	Cons
Fly.io GPU instance (A100 80GB)	a100-80gb shared, 24h	£550-700	Same provider as fly app; close to disk; familiar ops	Limited availability, pricing changes, single-region
RunPod / Lambda Labs (rented A100)	A100 80GB on-demand	£400-600	Cheapest GPU rental currently, hot-swap regions	Vendor lock-in; SLA varies
Dedicated colo / on-prem GPU host	2× A6000 48GB (~70B fits)	£1,200-2,500 (incl rack/power)	Full sovereignty; no cloud egress; audit-friendly for defence	Capex / setup cost; ops overhead
Hybrid: Fly inference + colo training	Fly for serve, colo for periodic re-train	£600-900	Best of both	More moving parts

Recommendation for v0.1: RunPod A100 80GB on-demand for first 3 months (£400-600/mo), evaluate dedicated colo at v0.2.

Inference stack: - **vLLM** as the serving layer (PagedAttention, continuous batching, speculative decoding for latency) - **OpenAI-compatible API** so the existing aria.llm.fallback chain swaps to ARIA-LLM with one config change (`ARIA_LLM_URL= ...`) - **Hot-fail to Anthropic** if ARIA-LLM container crashes (10-20s health check window) - **Streaming SSE** native — matches existing chat/stream endpoint - **Prompt caching** in vLLM for the 23-clause constitution (cached across all requests; 60-80% prompt token cost reduction internally)

Migration sequence: 1. ARIA-LLM serves Tier 0 + Tier 1 (student loops, research, autonomous tasks) for 1 week. Monitor: error rate, latency, output quality 2. Promote ARIA-LLM to Tier 2 (DD layers 1-4) for 1 week. Monitor as above. 3. Promote ARIA-LLM to Tier 3 (Layer 5c commercial coherence) for 1 week. A/B test against DeepSeek baseline. 4. Promote ARIA-LLM to Tier 4 (chat / audit-grade DD) — Anthropic demotes to break-glass fallback only. 5. Total migration window: 4 weeks.

Phase 4 exit criteria: 1. ARIA-LLM serves $\geq 95\%$ of production LLM calls 2. Customer-facing latency $p50 \leq 4s$, $p95 \leq 12s$ 3. Adversarial pass rate stays $\geq 90\%$ across migration 4. No customer-reported regression in DD output quality 5. Anthropic spend drops to $<£20/\text{mo}$ (break-glass only)

Operator decisions for Phase 4: - Approve £400-600/mo GPU rental ongoing budget - Approve break-glass policy (when does Anthropic fire? what's the rate-limit threshold?) - Approve customer communication on the migration (Pro Intel customers should know: "ARIA's underlying model is now sovereign")

Phase 5 — Full Independence + ARIA-LLM v1.0 (Month 5-9)

Goal: ARIA-LLM is the model. Anthropic / DeepSeek / Groq are removed from the codebase entirely. ARIA-LLM v1.0 specialised for security and defence is the product.

v1.0 differentiators vs v0.1: - **Larger fine-tune corpus:** 25k-50k SFT pairs (5x v0.1) — accumulated through 3-6 months of continuous harvest - **Multi-language:** Portuguese (Lusophone moat), French (West Africa), Arabic (Gulf + MENA), Spanish (LatAm) — fine-tuned on defence-DD content in each language. Achieves at-least 80% of the English performance in each. - **Tool use:** native function-calling fine-tune — ARIA-LLM produces structured JSON tool calls without prompt engineering. Maps directly to the 12 DD pipeline endpoints (R-F66..R-F84). - **Citation discipline baked in:** trained to emit inline citations with URL + retrieved-at timestamp by default; not via prompt instruction. - **Constitution baked in:** trained on constitutional-violation examples + correct refusals. The 23-clause check passes $\geq 99\%$ of outputs without runtime gating. - **Reasoning chain:** produces visible reasoning in `<thinking>` tags natively (like Anthropic's extended thinking). Operator can audit reasoning before relying on conclusions.

Phase 5 deliverables: - ARIA-LLM v1.0 weights (Arkmurus-controlled storage) - Open release decision: keep proprietary (defence-DD competitive moat) OR open-weight under Apache-2.0 (positions Arkmurus as the defence-DD AI standard) - Audit-grade evaluation report: ARIA-LLM v1.0 vs every leading model on a 1,000-question defence-DD benchmark - Customer migration: every Pro Intel customer sees ARIA-LLM by default; no code path to external LLM in customer-facing flows

Phase 5 exit criteria (ARIA is independent): 1. Anthropic / DeepSeek / Groq removed from `aria.llm.fallback` chain 2. Air-gap test passes: disconnect Fly.io from external internet for 1 hour; ARIA continues to serve chat / DD / autonomous tasks correctly using only ARIA-LLM 3. Defence-DD benchmark: ARIA-LLM v1.0 outperforms GPT-4 / Claude 3.5 Sonnet on the held-out defence-DD eval set 4. Constitutional adherence: $\geq 99\%$ of outputs pass the 23-clause check without runtime gating 5. Pro Intel customer NPS does not regress through the migration

Operator decisions for Phase 5: - Open vs proprietary licensing of ARIA-LLM v1.0 - Customer communication on the v1.0 launch (this is a product positioning event — "the only sovereign AI for defence-DD") - Whether to retain Anthropic / DeepSeek / Groq as break-glass configuration options or remove entirely

4. Best-in-Class in Security and Defence — What Makes ARIA-LLM Unique

A general-purpose LLM is vast and shallow. ARIA-LLM is narrow and deep — this is the correct architecture for the defence-DD use case.

The depth advantages:

1. **Constitutional grounding:** every output is constrained by the 23-clause constitution baked into training. No general LLM has defence-DD-specific output discipline at the weights level.
2. **Citation-first output:** trained to emit inline citations with retrieval timestamps. A general LLM has to be prompted into citing sources; ARIA-LLM does it natively.
3. **Sanctions-first reasoning:** trained that sanctions screening precedes commercial assessment (Clause 7). Ordering is structural, not prompted.
4. **Multi-jurisdictional sanctions awareness:** trained on the 28-list dataset_id mapping (R-F68). Knows OFAC SDN $\neq$ EU FSF $\neq$ HMT OFSI implicitly.
5. **FATF typology pattern recognition:** trained on the encoded typology library (R-F72) — recognises shell-company / TBML / VA / intermediary patterns from prose without explicit framework prompts.
6. **ECCN / Wassenaar fluency:** trained on the 105-system catalogue (R-F54) + the export-control regime corpus. Maps a weapon description to its ECCN classification natively.
7. **Defence-DD vocabulary depth:** ITAR vs EAR vs Wassenaar; CMC vs 1260H vs Entity List; UoR vs OTA vs FAR vs DFARS — all native vocabulary, not prompted glossary.
8. **Adversarial robustness baked in:** trained on the 33-attack library (R-F59 + R-F80) preference pairs. The base model's refusal patterns are reinforced specifically for defence-DD social engineering.
9. **Counter-intelligence awareness:** trained on the R-F84 patterns. Recognises reputation-washing in input documents during DD ingest.

10. **Persona discipline:** the six sectors (broker / oem_export / government_acquisition / compliance / banking_insurance / journalist) are trained as switchable personas with sector- specific vocabulary and reasoning posture.

A general-purpose LLM would need every one of these to be prompt- engineered into every call. ARIA-LLM has them in the weights. That is the moat.

5. Cost Summary

Phase	One-time	Monthly ongoing	Saves vs Anthropic-only
Phase 1 — Tiered substitution	£25 (Fly machine bump)	£90-130 (vs ~£385 today)	~£250-300/mo from month 1
Phase 2 — Corpus accumulation	£0 (uses existing infra)	Same as Phase 1	—
Phase 3 — v0.1 Fine-tune	£3,000-4,500 (compute)	Same as Phase 2	—
Phase 4 — Production deploy	£0 (rented GPU)	£400-600 (GPU rental)	~£60-90/mo <i>less</i> than Phase 1 (smaller external LLM bill, larger GPU bill)
Phase 5 — v1.0 + independence	£4,000-6,000 (re-train)	£400-700 (GPU + minimal break-glass)	~£100-150/mo less than Phase 1

Total one-time cost across all phases: £7,000-10,500. Steady-state monthly cost in independence mode: £400-700. Anthropic-equivalent monthly cost at scale: £1,500-3,000+.

The ROI inflects in month 6-9 depending on customer growth.

6. Risk Register

Risk	Likelihood	Impact	Mitigation
ARIA-LLM v0.1 underperforms Anthropic on chat	Medium	High — customer-facing	Phase 4 migration is gated on adversarial pass rate $\geq 90\%$ and customer-eval rubric; Anthropic stays as break-glass through migration
Open-weight base model licence change (vendor reneges)	Low	High	Pin to a specific commit / weight checksum; have Llama → Qwen swap path ready
GPU rental price spike (datacenter pressure)	Medium	Medium	Multi-vendor: RunPod / Lambda Labs / Vast.ai / Fly. Hot-swap takes <1h

Risk	Likelihood	Impact	Mitigation
Training corpus PII leakage into weights	Low (R-F67 redacts)	High (regulatory)	Phase 2 legal review; per-domain quality threshold; LoRA-only fine-tune means base model isn't poisoned
Customer SLA regression during Phase 4 migration	Medium	Medium	Migration is staged across 4 weeks; per-tier rollout; A/B tests at each tier promotion
Inference latency degradation vs Anthropic	Low (vLLM is fast)	Medium	Speculative decoding + prompt caching + smaller draft model for Tier 0 calls
Adversarial robustness loss in fine-tune	Low	High	DPO on R-F59 + R-F80 preference pairs is part of training; held-out adversarial set tests every checkpoint
Lusophone / non-English defence-DD quality	Medium	Medium for Arkmurus's market	v0.2 multilingual fine-tune scheduled in Phase 5; v0.1 may have a 5-10pp quality gap on PT/AR/FR
Compute cost overrun on training	Medium	Low	LoRA training caps the cost ceiling; if we exceed budget we pause and re-baseline

7. Where R-F86 through R-F90 Fit

The numbered improvements proposed earlier in this session map to **Phase 1 + Phase 2** of this roadmap. They are the tactical pieces of strategic shifts:

R-F#	Roadmap location	Status
R-F86 SearXNG / free search	Phase 1 (immediate cost reduction)	Pending — start of next session
R-F87 Local LLM tier (Ollama 8B)	Phase 1 (Tier 0 substitution)	Pending — start of next session
R-F87a Model-tier router	Phase 1 (the dispatcher logic)	Pending — bundled with R-F87
R-F88 Learning-progress tracker	Phase 2 (corpus completeness measurement)	Pending
R-F89 Knowledge coverage heatmap	Phase 2 (gap visualisation)	Pending
R-F90 Continuous-update orchestrator	Phase 2 (max-staleness contract)	Pending
R-F91+ Corpus curation pipeline	Phase 2 (training data collection)	Designed, not built
R-F92+ Fine-tune harness	Phase 3 (the SFT + DPO machinery)	Designed, not built

R-F#	Roadmap location	Status
R-F93+ vLLM deployment	Phase 4 (production serving)	Future

8. The 30-Day Action Pack

This is what a useful next 30 days looks like, in priority order:

Week 1 (next session, 4-6 hours dev): - Ship R-F86 (SearXNG self-host) - Ship R-F87 (Ollama 8B + model-tier router) - Operator: rotate `ARIA_INTERNAL_TOKEN` (still pending), top up Brave (still pending), set `REPORT_SIGNING_KEY` - Operator: approve `ARIA_OUTPUT_HARVEST_ENABLED=1` after 3-7 day threshold validation

Week 2-3: - Ship R-F88 + R-F89 + R-F90 (learning tracker, coverage heatmap, continuous-update orchestrator) - Verify Phase 1 cost reduction in `/api/aria/cost/monthly` - Begin SFT corpus accumulation — verify daily harvest count

Week 4: - Operator: scope GPU rental (RunPod / Lambda Labs) - Operator: legal review of harvested corpus for fine-tune PII - Begin defence-DD evaluation set construction (500 held-out questions)

By **Day 30:** Phase 1 complete, Phase 2 substantially underway, Phase 3 prerequisites in motion.

9. The One-Sentence Strategic Anchor

ARIA's brain has been sovereign for months; the next nine months make her mind sovereign too. ARIA-LLM v1.0 is a 70-72B parameter open-weight model fine-tuned on Arkmurus's own captured defence-DD outputs, the 23-clause constitution, and the curated security and defence corpus, served on Arkmurus-controlled compute. At v1.0 she is independent; at v1.0 she is the LLM in defence-DD.

Generated 2026-05-09 EOD. Strategic anchor for the next 3-9 months. Companion to `system_assessment_2026_05_09_eod.md` (current state), `architecture_2026_05_09.md` (current architecture), and `recommendations_complete_2026_05_09.md` (today's improvements).